Located, Not Secured: Principled Limits of Interpretability-Based Control over Agent Actions

A synthesis of one positive lever and five limits, and the turn to auditing

Caio Vicentino
OpenInterpretability
Fortaleza, Brazil
ORCID: 0009-0003-4331-6259
caio@openinterp.org

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Abstract

A recurring hope in agent safety is that mechanistic interpretability will let us *control* an agent: read an internal state, intervene, and steer behavior. Across an arc of pre-registered studies on an open-weight reasoning agent (Qwen3.6-27B), we find a sharp and consistent picture. Interpretability *locates* a real, causal control surface — a late action-commitment band (∼L51–63) that, unlike the mid-layer "task-done" verdict, can elicit and *brake* irreversible actions, generalizing across six actions and three architectures. But the control it affords is *not securable*, via five limits we make precise: (1) **detect $\neq$ control** — the clean "done" feature predicts the stop (AUROC 0.91) yet clamping it does nothing ($\Delta P = -0.001$); (2) **felt $\neq$ granted** — the authorization direction reads the authorization the model *feels*, allowing 21/21 realistic over-reaches an external check catches; (3) **form $\neq$ granted** — a high-AUROC "authorization" direction (0.838) collapses to 0.08 under structure-matching, i.e. it read scaffold, not concept; (4) **control $\neq$ robust control** — the late brake collapses (attack success $0 \rightarrow 1.0$ at a small budget) under an adaptive white-box adversary; (5) **intervention is easy where unneeded** — in the sincere-error regime a strong reasoning model already self-corrects, so the intervention is unnecessary, while the regime where it is needed (adversarial) is exactly where it is fragile. The conclusion: interpretability locates where behavior is decided but does not secure it. The actionable implication is a regime split — use interpretability to *audit and monitor* a fixed model (non-adversarial, where it wins), not to *defend* against an adversary optimizing against a known locus.

1 Introduction

If an agent is about to do something harmful and we can *see* it internally, can we *stop* it? The appeal of interpretability-based control — refusal directions, representation rerouting, feature clamping, activation steering — rests on a yes. This paper synthesizes a pre-registered arc that answers: interpretability *locates* the control surface, and sometimes acts through it, but the control is not securable. We state one positive result and five principled limits, each from a separate study with the same causal-rigor protocol (random-feature baseline, control-token normalization, structural-rigidity sweep, structure-matched control, behavioral confirmation). The synthesis is not a counsel of despair: it is a redirection. Where interpretability *loses* (adversarial control) is not where it is most useful; where it *wins* (non-adversarial auditing) is, and a companion result—*The Late Channel* [8]—shows what that looks like.

2 Interpretability locates the control surface (the positive)

For a long-horizon coding agent, control of the termination decision lives not at the mid-layer "task is done" verdict but in a late action-commitment block ($\sim$30 layers downstream, L51–63): a task-matched late donor flips a real `finish` (42%, p=0.031) while the verdict layer is inert [1]. This late lever *generalizes* — finish→edit, and from eliciting to *braking* — and works as a brake on genuinely irreversible actions: a simulated `send_transaction` collapses 0.998 → 0.00 with 100% redirect to a safe action, replicating across six action domains and three architectures [2, 3]. A single late-layer "authorization direction" both detects *and* controls an agent's commitment to unauthorized irreversible actions [4]. **Interpretability locates a real, causal, generalizing control surface.** The rest of the paper is about what that surface does *not* give us.

3 Limit 1 — Detect ≠ control

The "task complete / all tests pass" feature is clean, interpretable, and predicts the `finish` action (AUROC 0.91). Yet clamping it to its success value at the decision point yields $\Delta P(\texttt{finish}) = -0.001$ (p=0.44) — identical to a random feature [5]. The model *represents* "done" as a named feature, but that is not what makes it stop. Detection is not control, even at the exact, named feature.

4 Limit 2 — Felt ≠ granted (blind to a class of error)

On a real filesystem agent, the authorization probe is operationally blind to the realistic model-origin error: it allows 21/21 over-reaches the agent makes in good faith (CI $[0.845, 1.0]$), while an external, task-grounded check catches all 21 [6]. The probe reads the authorization the model *feels*, not the one the user *granted*. An internal monitor inherits the model's own judgment error — so defense-in-depth is necessary, because the internal and external failure modes are disjoint.

5 Limit 3 — Form ≠ granted (confoundable)

A frozen "authorization" direction with AUROC 0.838 collapses to 0.08 under structure-matching: it had separated on conversation structure, lexicon, and punctuation, not the concept [7]. A high cross-context AUROC without a named, structure-controlled feature is not evidence of a concept — a lesson that became a mandatory naming + structure-matched gate in the protocol.

6 Limit 4 — Control ≠ robust control

The same late action-commitment brake that suppresses `send_transaction` (1.00 → 0.00) is reopened by an adaptive white-box adversary that knows the brake layer and donor: a budget-4 signed-Adam embedding-space attack takes attack success 0 → 1.0, while a norm-matched random perturbation does nothing (companion result, this arc). The locus that *controls* in the benign regime is not adversarially robust — mirroring the broader finding that representation- and detection-level defenses fall to adaptive attacks.

7 Limit 5 — Intervention is easy where unneeded

Finally, even granting a working detector, the act it triggers is caught in a bind. On 32 agent-trap scenarios, the strong reasoning model already chose the safe action without any chain-of-thought in 30/32 cases — in the *sincere-error* regime it self-corrects, so an intervention is unnecessary. The regime where intervention is needed is the adversarial one (Limit 4), where mechanical injection collapses and a behavioral re-prompt can be rationalized. Reliable action exists where it is least needed; where the error is real, no intervention is robust.

8 Synthesis: locating is not securing

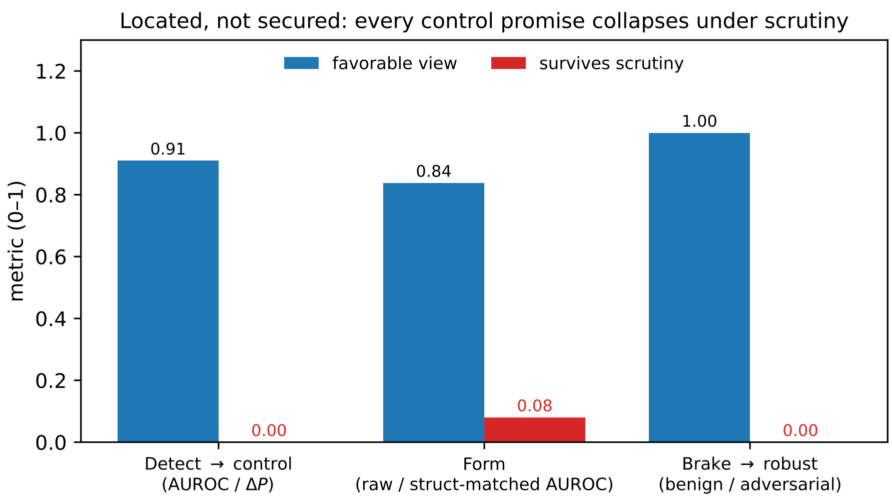

Figure 1: Three of the five limits, as matched pairs with identical semantics — a favorable interpretability metric (blue) vs what survives the right control (red): detection AUROC 0.91 vs control effect ≈ 0; raw AUROC 0.838 vs structure-matched 0.08; benign brake suppression vs adversarial robustness ($1 - \text{ASR}$). The other two limits (felt $\neq$ granted, intervention easy where unneeded) concern an internal monitor's blind spot and are qualitative — stated in the text and Table 1, not plotted.

	What interpretability gives	The limit
Locate the lever	late action band, causal, generalizes	— (this part works)
Detect the state	"done" feature, AUROC 0.91	clamping does nothing ($\Delta P = -0.001$)
Read authorization	a late direction	reads *felt*, allows 21/21 real over-reaches
Trust the AUROC	0.838 separability	0.08 under structure-matching
Brake the action	$1.0 \to 0.0$ suppression	$0 \to 1.0$ under adaptive attack
Act on the trigger	inject or re-prompt	easy where unneeded, fragile where needed

Table 1: One positive lever, five limits. Control power rises toward the late action channel; security guarantees do not follow it.

Locating *where* behavior is decided is necessary but nowhere near sufficient for *securing* it.

The limits are orthogonal: error-blindness, confoundability, non-robustness, and the easy-where-unneeded bind are different holes, none closed by a better localization.

9 Implication: a regime split

Why does interpretability-based control lose here? The adversarial-control regime lets an optimizer move activations against a known locus; activation malleability then makes it a losing game by construction. But the same internal access *wins* in the non-adversarial *auditing* regime — a fixed model examined as a scientific object, with no optimizer fighting the probe. "Detect $\neq$ control" stops mattering when the goal is detect/understand, not control-under-attack. The companion result *The Late Channel* [8] shows this concretely: the same late band that fails as a robust control point is a readable, causal locus for *monitoring* a reasoning agent. The recommendation that falls out of the whole arc: use interpretability to audit and monitor, not to defend.

10 Methods (the through-line)

Every result above passed one causal-rigor protocol: random-feature and shuffled-label baselines, control-token normalization for steering, a structural-rigidity α-sweep, a structure-matched control and feature naming, behavioral confirmation (a real generation, not a logit), and paired exact tests. This shared discipline is what lets five separate negatives be read as one principled picture rather than a string of failures.

11 Limitations

Simulated decision points; white-box interventions; a single model family (with two cross-architecture checks); modest n in several studies; the adversarial attacks use a strong continuous-embedding threat model whose deployment realism is contested. The positive (the located lever) is the smallest-n result; the load-bearing claims here are the limits, which are robust.

References

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